

Processando config 0, seed 1, teste: Rabbit  
 xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx  
 Starting Rabbit: nb = 16777216  
 Version: TestU01 1.2.3  
 xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx

\*\*\*\*\*  
 HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
 Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

smultin\_MultinomialBitsOver test:

-----  
 N = 2, n = 8388608, r = 0, s = 32, L = 38,  
 Sparse = TRUE  
  
 Number of bits = n = 8388608  
 Number of cells =  $2^L$  = 274877906944  
 Expected number per cell =  $1 / 32768$   
 EColl =  $n^2 / (2k)$  = 128  
 Hashing = TRUE  
  
 Collision test  
 CollisionOver: density =  $n / k = 1 / 32768.00$   
 Expected number of collisions =  $\mu = 128.00$

-----  
 Results of CollisionOver test:

POISSON approximation :  
 Expected number of collisions =  $N \cdot \mu$  : 256.00  
 Observed number of collisions : 529  
 p-value of test : 2.2e-50 \*\*\*\*\*

-----  
 Total number of cells containing j balls

|       |   |              |
|-------|---|--------------|
| j = 0 | : | 549739037201 |
| j = 1 | : | 16776158     |
| j = 2 | : | 529          |
| j = 3 | : | 0            |
| j = 4 | : | 0            |
| j = 5 | : | 0            |

-----  
 CPU time used : 00:00:04.58

Generator state:  
 16777216 bits have been read.

\*\*\*\*\*  
 HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
 Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

snpair\_ClosePairsBitMatch test:

-----  
N = 1, n = 262144, r = 0, t = 2

-----  
Max of all bit distances : 26.00  
p-value of test : 1.9e-6 \*\*\*\*\*

-----  
CPU time used : 00:00:00.05

Generator state:  
33554432 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

snpair\_ClosePairsBitMatch test:

-----  
N = 1, n = 131072, r = 0, t = 4

-----  
Max of all bit distances : 15.00  
p-value of test : 4.7e-10 \*\*\*\*\*

-----  
CPU time used : 00:00:00.07

Generator state:  
50331648 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

svaria\_AppearanceSpacings test:

-----  
N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of  $n = (K + Q)L = 15728640$  bits  
Q = 524288 initialization blocks  
K = 524288 blocks for the test  
the blocks have L = 15 bits

-----  
Normal statistic : -0.47  
p-value of test : 0.68

-----  
CPU time used : 00:00:00.00

Generator state:  
67108864 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

scomp\_LinearComp test:

-----  
N = 1, n = 18176, r = 0, s = 32

-----  
Number of degrees of freedom : 7  
Chi2 statistic for size of jumps : 2.60  
p-value of test : 0.92

-----  
Normal statistic for number of jumps : -84.81  
p-value of test : 1 - eps1 \*\*\*\*\*

-----  
CPU time used : 00:00:00.00

Generator state:  
67127040 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

scomp\_LempelZiv test:

-----  
N = 1, n = 16777216, r = 0, s = 32, k = 24

-----  
Normal statistic : 0.16  
p-value of test : 0.44

-----  
CPU time used : 00:00:00.41

Generator state:

83904256 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

sspectral\_Fourier1 test:

-----

N = 1, n = 1048576, r = 0, s = 32, k = 20

-----

Normal statistic : -0.74  
p-value of test : 0.77

-----

CPU time used : 00:00:00.21

Generator state:

84952832 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

sspectral\_Fourier3 test:

-----

N = 8065, n = 2048, r = 0, s = 32, k = 11

-----

Kolmogorov-Smirnov+ statistic = D+ : 0.021  
p-value of test : 0.63

Kolmogorov-Smirnov- statistic = D- : 0.022  
p-value of test : 0.61

Anderson-Darling statistic = A2 : 0.23  
p-value of test : 0.98

-----

CPU time used : 00:00:00.11

Generator state:

101728032 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

sstring\_LongestHeadRun test:

-----  
N = 1, n = 60, r = 0, s = 32, L = 279616

-----  
Number of degrees of freedom : 3  
Chi-square statistic : 3.00  
p-value of test : 0.39

-----  
Global longest run of 1 : 26.00  
p-value of test : 0.12

-----  
CPU time used : 00:00:00.06

Generator state:  
118504992 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

sstring\_PeriodsInStrings test:

-----  
N = 1, n = 524288, r = 0, s = 31

-----  
Number of degrees of freedom : 94  
Chi-square statistic : 109.73  
p-value of test : 0.13

-----  
CPU time used : 00:00:00.01

Generator state:  
135282208 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

sstring\_HammingWeight test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Number of degrees of freedom : 22  
Chi-square statistic : 24.89  
p-value of test : 0.30  
-----

-----  
CPU time used : 00:00:00.01

Generator state:  
152059424 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

sstring\_HammingCorr test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : -0.41  
p-value of test : 0.66  
-----

-----  
CPU time used : 00:00:00.00

Generator state:  
168836640 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

sstring\_HammingCorr test:

-----  
N = 1, n = 262144, r = 0, s = 32, L = 64

-----  
Normal statistic : -1.16  
p-value of test : 0.88  
-----

-----  
CPU time used : 00:00:00.00

Generator state:  
185613856 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

sstring\_HammingCorr test:

-----

N = 1, n = 131072, r = 0, s = 32, L = 128

-----  
Normal statistic : -1.47  
p-value of test : 0.93

-----  
CPU time used : 00:00:00.00

Generator state:  
202391072 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 174.92  
p-value of test : 0.53

-----  
CPU time used : 00:00:00.01

Generator state:  
219168288 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 3767.12  
p-value of test : eps \*\*\*\*\*

-----  
CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 489  
Chi-square statistic : 2277.14  
p-value of test : eps \*\*\*\*\*

-----  
CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

sstring\_AutoCor test:

-----  
N = 1, n = 16777153, r = 0, s = 32, d = 1

```
-----
Normal statistic          :   -0.14
p-value of test          :    0.56
-----
```

```
CPU time used            :   00:00:00.00
```

```
Generator state:
 269499904 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config0_seed1.bin
```

```
sstring_AutoCor test:
```

```
-----
      N =  1,   n = 16777154,   r =  0,   s = 32,   d = 2
-----
```

```
-----
Normal statistic          :   -1.49
p-value of test          :    0.93
-----
```

```
CPU time used            :   00:00:00.01
```

```
Generator state:
 286277088 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config0_seed1.bin
```

```
sstring_Run test:
```

```
-----
      N =  1,   n = 3355443,   r =  0,   s =  32
-----
```

```
-----
Total number of 1 runs:  3355443
```

```
Number of degrees of freedom      :   36
Chi2 statistic for number of runs :   29.16
p-value of test                   :    0.78
-----
```

```
-----
Total number of bits:  13422368
```

```
Normal statistic for number of bits :    0.12
p-value of test                     :    0.45
```

-----  
CPU time used : 00:00:00.05

Generator state:  
299699456 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

smarsa\_MatrixRank test:

-----  
N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

-----  
Number of degrees of freedom : 3  
Chi-square statistic : 5.11  
p-value of test : 0.16

-----  
CPU time used : 00:00:00.06

Generator state:  
316476672 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

smarsa\_MatrixRank test:

-----  
N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

-----  
Number of degrees of freedom : 2  
Chi-square statistic : 2.82  
p-value of test : 0.24

-----  
CPU time used : 00:00:00.08

Generator state:  
333167872 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

-----  
Test on the values of the Statistic H

|                              |   |         |       |
|------------------------------|---|---------|-------|
| Number of degrees of freedom | : | 42      |       |
| ChiSquare statistic          | : | 2843.26 |       |
| p-value of test              | : | eps     | ***** |

-----  
Test on the values of the Statistic M

|                              |   |         |       |
|------------------------------|---|---------|-------|
| Number of degrees of freedom | : | 42      |       |
| ChiSquare statistic          | : | 1223.68 |       |
| p-value of test              | : | eps     | ***** |

-----  
Test on the values of the Statistic J

|                              |   |        |       |
|------------------------------|---|--------|-------|
| Number of degrees of freedom | : | 64     |       |
| ChiSquare statistic          | : | 284.38 |       |
| p-value of test              | : | eps    | ***** |

-----  
Test on the values of the Statistic R

|                              |   |       |  |
|------------------------------|---|-------|--|
| Number of degrees of freedom | : | 36    |  |
| ChiSquare statistic          | : | 55.67 |  |
| p-value of test              | : | 0.02  |  |

-----  
Test on the values of the Statistic C

|                              |   |       |  |
|------------------------------|---|-------|--|
| Number of degrees of freedom | : | 22    |  |
| ChiSquare statistic          | : | 28.09 |  |
| p-value of test              | : | 0.17  |  |

-----  
CPU time used : 00:00:00.06

Generator state:  
349945088 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

-----  
Test on the values of the Statistic H

|                              |   |         |       |
|------------------------------|---|---------|-------|
| Number of degrees of freedom | : | 94      |       |
| ChiSquare statistic          | : | 215.04  |       |
| p-value of test              | : | 1.8e-11 | ***** |

-----  
Test on the values of the Statistic M

|                              |   |         |       |
|------------------------------|---|---------|-------|
| Number of degrees of freedom | : | 94      |       |
| ChiSquare statistic          | : | 227.07  |       |
| p-value of test              | : | 5.1e-13 | ***** |

-----  
Test on the values of the Statistic J

|                              |   |        |  |
|------------------------------|---|--------|--|
| Number of degrees of freedom | : | 512    |  |
| ChiSquare statistic          | : | 561.50 |  |
| p-value of test              | : | 0.06   |  |

-----  
Test on the values of the Statistic R

|                              |   |       |  |
|------------------------------|---|-------|--|
| Number of degrees of freedom | : | 90    |  |
| ChiSquare statistic          | : | 93.19 |  |
| p-value of test              | : | 0.39  |  |

-----  
Test on the values of the Statistic C

|                              |   |       |  |
|------------------------------|---|-------|--|
| Number of degrees of freedom | : | 50    |  |
| ChiSquare statistic          | : | 67.72 |  |
| p-value of test              | : | 0.05  |  |

-----  
CPU time used : 00:00:00.03

Generator state:

366722304 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed1.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

-----  
Test on the values of the Statistic H

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 130    |
| ChiSquare statistic          | : | 128.14 |
| p-value of test              | : | 0.53   |

-----  
Test on the values of the Statistic M

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 131    |
| ChiSquare statistic          | : | 153.00 |
| p-value of test              | : | 0.09   |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 161    |
| ChiSquare statistic          | : | 173.46 |
| p-value of test              | : | 0.24   |

-----  
Test on the values of the Statistic R

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 130    |
| ChiSquare statistic          | : | 133.48 |
| p-value of test              | : | 0.40   |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 89    |
| ChiSquare statistic          | : | 77.87 |
| p-value of test              | : | 0.79  |

-----  
CPU time used : 00:00:00.03

Generator state:

383499104 bits have been read.

===== Summary results of Rabbit =====

|                       |                     |
|-----------------------|---------------------|
| Version:              | TestU01 1.2.3       |
| Generator:            | ufile_CreateReadBin |
| Number of bits:       | 16777216            |
| Number of statistics: | 39                  |

Total CPU time: 00:00:06.05  
 The following tests gave p-values outside [0.001, 0.9990]:  
 (eps means a value < 1.0e-300):  
 (eps1 means a value < 1.0e-15):

|    | Test                      | p-value  |
|----|---------------------------|----------|
| 1  | MultinomialBitsOver       | 2.2e-50  |
| 2  | ClosePairsBitMatch, t = 2 | 1.9e-6   |
| 3  | ClosePairsBitMatch, t = 4 | 4.7e-10  |
| 5  | LinearComp                | 1 - eps1 |
| 16 | HammingIndep, L = 32      | eps      |
| 17 | HammingIndep, L = 64      | eps      |
| 24 | RandomWalk1 H             | eps      |
| 24 | RandomWalk1 M             | eps      |
| 24 | RandomWalk1 J             | eps      |
| 25 | RandomWalk1 H (L = 1024)  | 1.8e-11  |
| 25 | RandomWalk1 M (L = 1024)  | 5.1e-13  |

-----  
 All other tests were passed